quirq infra (qq) · v0

# How to use quirq infra

A Chromium-style CI/CD system for every quirq repo: the parts, how they fit together, how you work with it, and how it is named and governed.

For engineers and agents new to quirq

Status refreshed 2026-10-05

WHAT QQ IS

# One config repo decides; small tools apply it

**Modelled on Chromium.** infra-config declares every repo, builder, gate rule and channel as TOML. The tools that apply policy read it at a pinned commit.

**Repo-agnostic.** Core tools never name a language or product; a product repo writes one manifest and gets the rest generated.

**Agents first.** Policy allows agents to land clean reverts, dependency rolls and docs alone. In v0 none of these land on their own (auto_land_repos is empty and roll auto-land is off). Canary promotion is the one thing agents do alone today.

**Public, GitHub-hosted.** 13 infra repos, all Apache-2.0, serving two product repos: xo-space and innernet.

STATUS · 2026-10-05

**Built.** v0 code merged in all 13 repos except GAT-03 (waits on owners), CFG-05 (partial) and suraj's items (CFG-04, ORG-02, ORG-04); repo rulesets applied (gate 6610664); suraj named owner of the org and every infra repo.

**Live:** generated CI on both repos, lkgr at both mains, perf records, tree status open; first canary shipped (started by hand).

**Not yet proven:** a red PR refused, 7 unattended canaries, an automatic revert.

PHASE v0

# v0: A base version of every repo, end to end on GitHub

Code merged except GAT-03, CFG-05 and suraj's items; exit test not yet passed

## GOAL

All 13 infra repos exist, are public, sit behind their own gate and do a thin first version of their job. A product change is built from its manifest, gated on the exact merge result, watched after it lands and shipped to canary by agents daily.

## SCOPE

Python 3.14 and Next.js 16 kinds; xo-space and innernet; GitHub merge queue + Actions behind an executor interface; only the canary channel live.

## EXIT TEST · AS OF 2026-10-05

1. 13 public repos, each with CODEOWNERS, a generated presubmit and a merge queue
   partly: all have CODEOWNERS; repo rulesets applied at gate 6610664 (per-repo coverage not verified); infra presubmits hand-written
2. Both product repos gated from their manifests; a red PR is refused
   builders required but on interim commands, not recipes adapters (open v0 work); no red-PR refusal yet
3. 7 daily canaries, no human touch; bad canary held; rollback under 10 min
   ran once, by hand (2026-10-05); unattended streak not started
4. A planted breaking commit is auto-reverted within caps, with a failure record
   waits on the gardener App (not yet created)
5. Scorecard v0 computed by a script
   test-pipelines scorecard workflow runs

PHASE v1 (plan)

# v1: Unattended canary, dev channel, fuzzing, failure tracking

Not started (follows v0, by inference)

## GOAL

The daily canary runs with nobody watching: it soaks, rolls itself back, fuzzes, files failure records and drafts postmortems. Dev opens with human approval. Flakes are handled, builds are hermetic and cached, and containers and docs repos join.

## SCOPE

Headline items: canary soak + auto rollback, dev channel, Python/Node/API fuzzing, flake quarantine, shared REAPI cache, PostHog as a health signal, docs and wiki onboarded, one-command idempotent setup.

## EXIT TEST

1. 14 green daily canaries in a row, zero human touches
2. A planted bad canary is rolled back in its soak window, with failure record and cited postmortem draft
3. A planted fuzz crash becomes one deduplicated issue that closes only when replay passes
4. Dev promoted 4 times, each by a human owner approving agent evidence
5. A failure class seen twice in 14 days yields a new test, fuzz target or gate
6. Flake rate weekly; cache hit rate at least 90%; main red under 60 min a week for 4 weeks

PHASE v2 (plan)

# v2: Stable, speed, depth and Launchpad

Not started (follows v1, by inference); stable and installers P4

## GOAL

Everything that runs on GitHub can also run on Launchpad, quirq's own cloud. Remote execution and perf detection reach Chromium's shape, fuzzing gains depth, and agents hold the gardener, roller and release rotations.

## SCOPE

Headline items: Launchpad backend and a gate without GitHub's merge queue, remote execution and autoscaled workers, cross-repo gating and rolls, perf bisection, stable promotion and staged rollout.

## EXIT TEST

1. The same build runs on github and launchpad with matching digests, chosen only by config
2. Gate p50 under 10 min and p90 under 20 min for 4 weeks
3. A planted 10% perf regression is alerted and bisected to its culprit
4. 8 weeks under suraj's intervention target, revert rate no worse than v1, policy changes owner-approved
5. P4: one stable promotion, a rollback drill under 10 min, xo-space installs follow a channel

COMPONENTS

# 13 infra repos, each one job

| Repo | What it does | Chromium part |
|---|---|---|
| `infra-config` | All policy as TOML; qqcfg validates, generates, delivers | infra/config + lucicfg |
| `depot` | The qq CLI: sync, build, test, upload, try, land, status | depot_tools |
| `sync` | Manifest schema quirq-repo/1 and its only parser, qqsync | gclient / DEPS |
| `recipes` | One adapter per target kind (python-service, pytest, node-app) | recipes |
| `toolchains` | Builds CPython 3.14, Node 24 + pnpm; publishes to ghcr by digest | CIPD / 3pp |
| `remote-build` | Executor interface (local, github) and an action cache | RBE / REAPI |
| `test-pipelines` | JUnit sink, write-once results, verdicts, failure records | ResultDB, LUCI Analysis |
| `gate` | Required checks from config; rulesets and merge queue as code | LUCI CV |
| `gardener` | Tree status, failure grouping, bisection, capped auto-revert | Sheriff-o-Matic, Bisection |
| `rollers` | Dependabot config + land check; toolchain pin roller | AutoRoll |
| `release` | lkgr, channel pointers, rollback, daily canary | lkgr finder, promote.py |
| `installer` | Clients follow a channel instead of main | Omaha |
| `perf` | Build size and benchmarks per commit, a record never a gate | perf dashboard |

REPO 1 OF 13 · CHROMIUM: infra/config + lucicfg

# infra-config · All policy as code

## WHAT IT DOES

- Declares every repo, builder, gate rule, channel and cap as TOML under config/
- qqcfg validate checks schemas, references and invariants config cannot loosen
- qqcfg generate writes the builders; qqcfg deliver writes them into a product checkout
- Every tool reads it at a pinned commit, so a change reaches a tool only after a pin bump

## v0 · AS OF 2026-10-05

- **Merged:** CFG-01..03, ORG-01; CFG-05 partly
- **main:** `310e326`
- **Live:** Generated builders on both repos, sink 28f49a0 (xo-space #220, innernet #44 merged); suraj owns org and 13 repos
- **Open:** suraj's values (CFG-04), other area owners (ORG-02), compute ceiling (ORG-04)

## YOU MEET IT AS

- `qqcfg validate [--todos]`
- `qqcfg generate`
- `qqcfg deliver <repo> <checkout>`
- `config/repos.toml, pipelines.toml, gate.toml`

## NEXT IN v1

- Dev channel on
- Health thresholds
- Fuzz schedules and filing caps
- Onboard docs, wiki, containers

REPO 2 OF 13 · CHROMIUM: depot_tools

# depot · The qq CLI

## WHAT IT DOES

- One command for developers and agents, pinned per repo in infra/repo.toml
- Fetches toolchains, builds and tests through recipes, uploads and lands changes
- qq try returns a run ID at once and delivers the verdict later, so agents never block
- An e2e-sync job, scheduled daily, checks that a fresh runner can sync both product repos

## YOU MEET IT AS

- `qq sync` · `qq fetch <url>`
- `qq build` · `qq test`
- `qq upload` · `qq try`
- `qq land` · `qq status (0 pass, 1 refused, 3 pending)`

## v0 · AS OF 2026-10-05

- **Merged:** DEP-01..04 and all audit fixes
- **main:** `741967c`
- **Live:** e2e-sync scheduled daily at 06:17 UTC for xo-space and innernet (runs not readable anonymously)
- **Open:** No depot v* tag can be cut until the release executor exists

## NEXT IN v1

- qq roll
- qq channel (hold, release, roll back)
- qq failure and qq postmortem

REPO 3 OF 13 · CHROMIUM: gclient / DEPS

# sync · The manifest and its only parser

## WHAT IT DOES

- Owns the manifest schema quirq-repo/1: qq version, toolchain pins by digest, typed targets
- qqsync is the one parser every other repo imports
- Pin check: every pin is a digest, and a fetch that does not match it fails
- A guard fails any other code that tries to parse the manifest itself

## YOU MEET IT AS

- `infra/repo.toml`
- `qqsync validate`
- `qqsync show`
- `qqsync guard`

## v0 · AS OF 2026-10-05

- **Merged:** SYN-01..04
- **main:** `d97e2f7`
- **Live:** Used by depot, recipes, gate, rollers, release and perf
- **Open:** none for v0

## NEXT IN v1

- Mirroring: builds pass with upstream registries blocked
- Schema quirq-repo/1.1: fuzz targets, probes, deploy targets

REPO 4 OF 13 · CHROMIUM: recipes

# recipes · One adapter per target kind

## WHAT IT DOES

- One adapter contract per kind: fetch, build, test, run, deploy, bench (+ package, fuzz)
- Adapters: python-service, pytest, node-app (Next.js); the loader finds them, no registry
- Property tests run inside test
- Deploys to a canary test environment for release's canary

## v0 · AS OF 2026-10-05

- **Merged:** REC-01..05 plus bench
- **main:** `db6ce0a`
- **Live:** Used by qq, release's canary and perf
- **Open:** Open v0 work: generated CI builders still run interim commands, not these adapters (exit test 2)

## YOU MEET IT AS

- `qq build / qq test call it`
- `release canary and perf call it`
- `adding a language = one adapter module`

## NEXT IN v1

- container-image and static-docs adapters
- Python, Node and API fuzzing
- Real canary deploy targets

REPO 5 OF 13 · CHROMIUM: CIPD / 3pp

# toolchains · Pinned, promoted toolchains

## WHAT IT DOES

- Builds CPython 3.14.8 and Node 24.21.0 with pnpm 11.28.2
- Publishes each build to public ghcr, addressed by digest
- A build is staged, then promoted by PR into promoted.toml
- A promotion-gate workflow runs on PRs and the merge queue; as a required repo check it is decided and turns on with the next settings run

## YOU MEET IT AS

- `promoted.toml`
- `toolchains.toml`
- `ghcr.io/quirq-ai/…@sha256`

## v0 · AS OF 2026-10-05

- **Merged:** TCH-01..03
- **main:** `44f8f95`
- **Live:** suraj is code owner of the gate paths (#14); promotion-gate runs on PRs and the queue (#15)
- **Open:** promotion-gate as a required check: decided; turns on with the next settings run (gate #25)

## NEXT IN v1

- Reproducibility check
- Container toolchain

REPO 6 OF 13 · CHROMIUM: RBE / REAPI (the goma slot)

# remote-build · Where actions run, and their cache

## WHAT IT DOES

- An executor interface with two backends: local and github
- A local action cache keyed by action digest, with fallback counters
- Launchpad plugs into the same seam later

## v0 · AS OF 2026-10-05

- **Merged:** RBE-01, RBE-02
- **main:** `62b2f04`
- **Live:** Interface and cache ready
- **Open:** qq does not call it yet

## YOU MEET IT AS

- `qqrbe exec/selftest --backend local|github`
- `recipes action shape`

## NEXT IN v1

- Shared REAPI cache (v2: remote execution, autoscaled workers)

REPO 7 OF 13 · CHROMIUM: ResultDB, LUCI Analysis

# test-pipelines · Results and verdicts

## WHAT IT DOES

- A sink in every builder stores each JUnit result in a write-once store
- Verdict: retry, then compare with the base commit
- Failure records with an issue mirror; gardener writes one per revert
- A scorecard workflow collects results, perf bundles and gate timing

## YOU MEET IT AS

- `sink step in each qq-*.yml`
- `qqresults`
- `scorecard workflow`

## v0 · AS OF 2026-10-05

- **Merged:** TST-01..04 and audit fixes
- **main:** `a642399`
- **Live:** Product results stored via sink 28f49a0, the v0 pin
- **Open:** Retry-then-compare is off in product builders

## NEXT IN v1

- Flake detection and quarantine with expiry
- Exoneration of known flakes
- New-test flakiness check
- Failure classes and recurrence

REPO 8 OF 13 · CHROMIUM: LUCI CV

# gate · What must pass before a change lands

## WHAT IT DOES

- Computes each repo's required checks from infra-config and its manifest
- Rulesets and the merge queue as code, applied by one command
- Agnosticism guard: no core file may name a language or tool
- Gate timing (queue to verdict) into the scorecard

## YOU MEET IT AS

- `qqgate (0 pass, 1 refused, 2 undecided, 3 not onboarded)`
- `settings/github.toml`
- `scripts/apply.sh one-liner`

## v0 · AS OF 2026-10-05

- **Merged:** GAT-01, 02, 04 and ORG-03
- **main:** `c372136`
- **Live:** Repo rulesets applied at 6610664; org rulesets off (GitHub Free)
- **Open:** GAT-03 waits on owners; promotion-gate required: decided; turns on with the next settings run

## NEXT IN v1

- Result reuse
- Admission bar enforcement
- Tree status in the gate

REPO 9 OF 13 · CHROMIUM: Sheriff-o-Matic, LUCI Bisection

# gardener · Keeps main green

## WHAT IT DOES

- Checks the tree every 5 minutes and publishes status per repo
- Groups failures, finds the culprit by bisection
- Opens clean revert PRs within the cap of 10 per 24 h
- Writes a failure record and postmortem stub per revert

## YOU MEET IT AS

- `tree-status branch: status/<repo>.json`
- `ledger branch (after a manual bootstrap run)`
- `auto_revert.toml caps`

## v0 · AS OF 2026-10-05

- **Merged:** GAR-01..04 and audit fixes
- **main:** `bf7d24d`
- **Live:** Tree status published; both repos open
- **Open:** Its GitHub App, not yet created (needed before exit test 4 can pass): until then it reports reverts but opens none

## NEXT IN v1

- Test-failure reverts
- Revert precision
- Postmortem drafting agent
- Canary bisection

REPO 10 OF 13 · CHROMIUM: AutoRoll

# rollers · Keeps dependencies and toolchains current

## WHAT IT DOES

- Ships Dependabot config to each product repo
- qq-roll-land checks that a roll is a clean patch or minor bump
- Rolls toolchain pins from promoted.toml as PRs from the quirqer App
- Weekly toolchain rotation

## YOU MEET IT AS

- `.github/dependabot.yml`
- `qq-roll-land.yml`
- branch `qq-roll/toolchains`

## v0 · AS OF 2026-10-05

- **Merged:** ROL-01, ROL-02 and lockfile tree check
- **main:** `4285aa4`
- **Live:** Dependabot config on both repos; first roll innernet #43 (TS 5→7, a major bump the land check does not treat as clean) merged by hand
- **Open:** Auto-land is off: a person merges every roll

## NEXT IN v1

- Weekly rolls for every pin (v2: cross-repo rolls with canary feedback)

REPO 11 OF 13 · CHROMIUM: lkgr finder, promote.py

# release · lkgr, channels and the daily canary

## WHAT IT DOES

- Moves lkgr to the newest all-green main commit every 10 minutes
- Channel pointers with digests, and rollback
- Daily canary at 06:17 UTC: build, test, deploy and probe, then move channels/canary
- Writes a daily canary report; a watchdog starts a missed canary, and since 2026-10-05 a 07:37 UTC routine dispatches lkgr, then the watchdog

## YOU MEET IT AS

- release-state branch
- channels.json (qq-channels/1)
- qq channel rollback

## v0 · AS OF 2026-10-05

- **Merged:** REL-01..04 and audit fixes
- **main:** 83ef917
- **Live:** lkgr at both repos' current main; first canary shipped both repos 2026-10-05 (started by hand) in release-state d08a2fd
- **Open:** The unattended 7-day streak; the release executor, needed for product-repo refs and, later, depot release tags (not for the canary)

## NEXT IN v1

- Canary soak and automatic rollback
- Canary fuzz stage
- Dev channel
- Escalation

REPO 12 OF 13 · CHROMIUM: Omaha

# installer · Installs follow a channel, not main

## WHAT IT DOES

- Reads release's channels.json for a repo and channel
- Resolves the commit and digest, then checks out and verifies exactly that
- v0 scope: xo-space test installs only, never real users' machines

## YOU MEET IT AS

- `qqinstall resolve --repo xo-space --channel canary`
- `qqinstall checkout · verify`
- `exit 0 ok, 1 mismatch, 2 error, 3 not published`

## v0 · AS OF 2026-10-05

- **Merged:** INS-01, INS-02 with audit fixes
- **main:** `d39f302`
- **Live:** Has a canary to resolve for both repos (since 2026-10-05)
- **Open:** Canary test machines are not set up yet

## NEXT IN v1

- Test installs follow dev (v2: xo-space follows a channel)

REPO 13 OF 13 · CHROMIUM: perf dashboard, Pinpoint

# perf · Performance on every commit

## WHAT IT DOES

- Records a benchmark per main commit for both repos, and innernet's build size
- Stores raw numbers with unit and runner type as test-pipelines bundles
- A record only in v0: never a gate or an alert

## YOU MEET IT AS

- perf-data branch
- infra-config config/perf.toml
- benchmarks: xo-space-server-start, innernet-search

## v0 · AS OF 2026-10-05

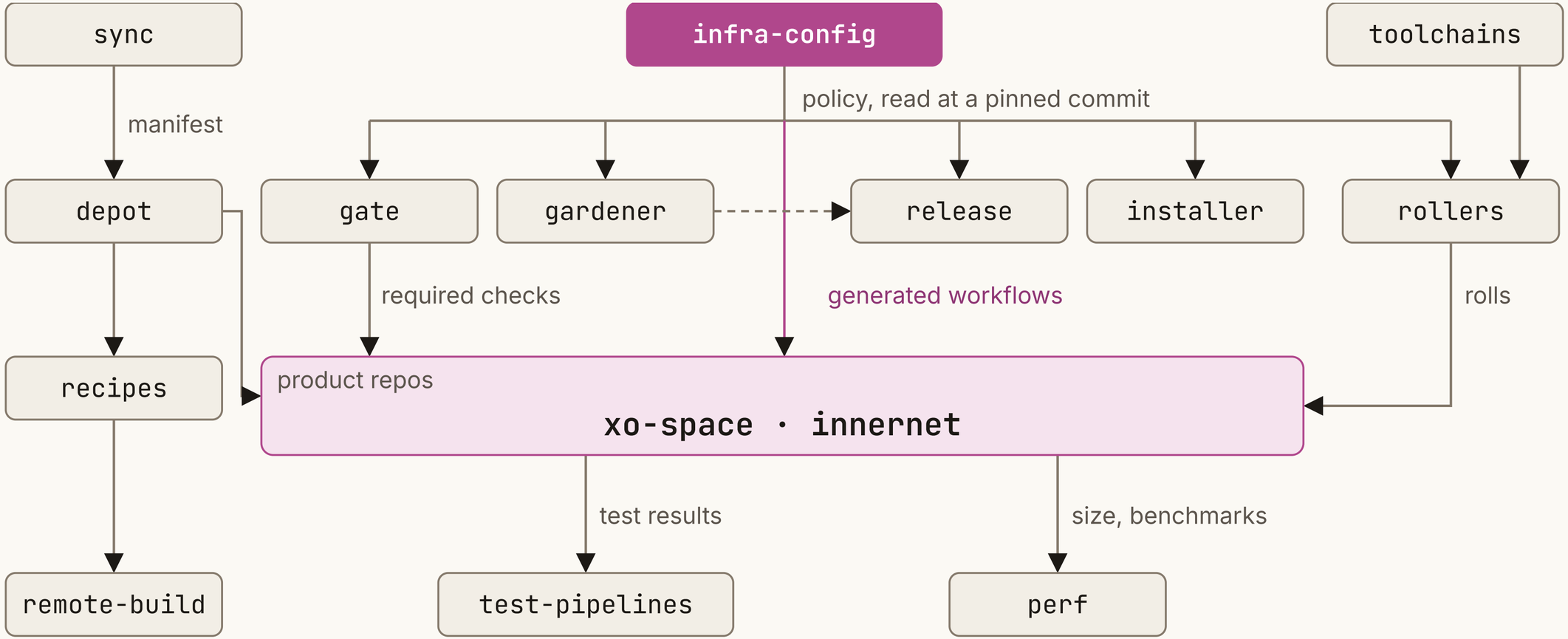
- **Merged:** PRF-01, PRF-02 and audit fixes
- **main:** d1d9765
- **Live:** Recording main commits on perf-data (newest record 2026-10-05 02:27 UTC)
- **Open:** One measurement per commit on shared runners, no baseline

## NEXT IN v1

- Trend and threshold alerts
- Bundle-size budget

HOW THEY INTERACT

# Policy flows down, results flow back




Dashed: gardener tells release which main commits are green, and release moves lkgr. Full list on the next slide.

HOW THEY INTERACT

# Every link is a pinned commit, never copied code

| From | To | What flows |
|---|---|---|
| `infra-config` | gate, gardener, release, installer, rollers | Policy via qqcfg at each tool's pin (pins.toml; rollers: infra-config.commit) |
| `infra-config` | product repos | Generated qq-*.yml workflows, written by qqcfg deliver into a PR |
| `sync` | depot, recipes, gate, rollers, release, perf | The manifest, read only through qqsync |
| `toolchains` | rollers, then product repos | promoted.toml digests, turned into pin-roll PRs |
| `depot` | recipes; gate | qq build and test run through recipes; qq land and status run the pinned qqgate |
| `recipes` | release, perf; remote-build | Canary and perf builds run through recipes; remote-build runs its actions (not yet called) |
| `gate` | product repos | Required checks and rulesets (repo rulesets applied at gate 6610664) |
| `product CI, perf` | test-pipelines | JUnit through the sink; perf result bundles; the scorecard |
| `test-pipelines` | gardener | Failing tests per run; gardener writes one failure record back per revert |
| `gardener` | release | What counts as a green post-submit (qqgarden), used to move lkgr |
| `release` | installer, depot | channels.json for installs; the qq channel rollback subcommand |

HOW TO USE IT · 1 OF 2

# From a fresh clone to an open PR

1

### Set up once

Clone depot and put its bin on PATH. In a product repo, qq runs the version pinned in infra/repo.toml (both pin 0.1.0, depot main).

2

### Get toolchains: `qq sync`

Fetches every pinned toolchain from public ghcr, checks its digest, links it under .qq/toolchains/. Runs daily for both repos.

3

### Build and test: `qq build`, `qq test`

Recipes run the targets and leave JUnit and results.json in .qq/out.

4

### Open a PR: `qq upload` or a plain push

The <repo>-presubmit check runs a drift check, then fetch, build and test, then stores every result in test-pipelines.

HOW TO USE IT · 2 OF 2

# From a verdict to a channel

`qq try` returns a run ID at once and delivers the verdict later to $QQ_HOME/verdicts/, so an agent never holds a call open.

`qq status` gives it on demand: exit 0 pass, 1 refused, 3 pending.

`qq land` queues the PR once the gate passes; the merge queue re-tests the exact merge result and squash-merges.

**After it lands**, the post-submit builder runs on every main commit and gardener checks the tree every 5 minutes, proposing a revert if main goes red.

**Dependencies** move by Dependabot PRs, checked by qq-roll-land, and toolchain roll PRs from quirqer (the App exists; none needed yet). A person merges every roll in v0.

**Ship.** release moves lkgr to the newest all-green main commit; the daily canary at 06:17 UTC builds, tests, deploys and probes it before moving channels/canary.

`qqinstall resolve --repo xo-space --channel canary` gives the commit and digest to install; both repos now have a canary to resolve.

**Measure.** perf records build size and a benchmark per main commit.

> Today: gate's repo rulesets (merge queue, required checks) were applied at gate 6610664. The first canary shipped both repos at 06:59 UTC on 2026-10-05, started by hand after GitHub never fired the 06:17 slot; a 07:37 UTC backstop now dispatches lkgr and the watchdog. Reverts wait on the gardener App (needed before exit test 4 can pass); product-repo git refs wait on the release executor.

WHAT A PRODUCT REPO GETS

# Write one manifest; the rest is generated

| What | Lands as | Live today |
|---|---|---|
| Manifest: qq version, toolchain pins by digest, targets | `infra/repo.toml` | both repos |
| Presubmit builder, the required check | `qq—<repo>—presubmit.yml` | both repos |
| Post-submit builder on every main push | `qq—<repo>—postsubmit.yml` | both repos |
| Drift check against infra-config | a step in each qq-*.yml; org qq-drift | step yes; org rule off (Free plan) |
| Merge queue, required checks, squash only | the qq-main ruleset (gate) | applied (gate 6610664) |
| Every test result stored | a sink step in each builder | yes |
| Tree status, culprits, revert PRs | gardener's tree-status branch | status yes; reverts need the gardener App |
| Dependabot + land check; toolchain rolls | dependabot.yml, qq-roll-land.yml | yes, auto-land off |
| Build size and benchmark per commit | perf's perf-data branch | yes |
| lkgr, daily canary, channels/canary | release-state record; refs later | lkgr yes; canary ran once, by hand |
| Test installs follow canary | qqinstall checkout and verify | canary to resolve; no test machines yet |

POLICIES

# Every policy is a TOML file in infra-config/config

| File | Decides · value today |
|---|---|
| `gate.toml` | Who may land what (declared): reverts, rolls and docs need a green gate; code needs an owner; policy needs suraj |
| `gate.toml` | Squash merges; a blocking builder needs p50 under 15 min and must finish under 40 |
| `pipelines.toml` | Builders per repo: blocking presubmit, post-submit mirror, canary deploy (run by release) |
| `kinds.toml` | Toolchains and target kinds: Python 3.14, Node 24 with pnpm |
| `flakes.toml` | Retries and flakes, read by test-pipelines' verdict |
| `auto_revert.toml` | At most 10 reverts per rolling 24 h; auto_land_repos empty, so every revert is proposed |
| `rollers.toml` | Dependabot for pip and npm; weekly toolchain rotation |
| `channels.toml, health.toml` | canary: agents, daily; dev: a human owner after 24 h soak and 3 green canaries; stable: suraj |
| `repos.toml` | Every repo: source, kinds, manifest path, channels, owners, deploy target |
| `gate settings/github.toml` | Rulesets as code: qq-main (merge queue, checks), ref and tag locks; repo rulesets applied at 6610664; org rulesets and Dependabot rules off |

Rules config cannot loosen live in tools/qqcfg.py: stable needs suraj, canary is agents-only, at most 10 auto-reverts a day, no secrets for builders that run PR code, public repos only.

**Approvals.** Today this is advisory: the qq-main ruleset requires 0 approvals, and code-owner review is enforced only on toolchains (gate settings/github.toml). The change-class approvals in gate.toml are declared, not enforced, until V0-GAT-03.

POLICIES

# How a policy change reaches a repo

### 1 · EDIT

Change config/<area>.toml in a PR. CI runs qqcfg validate: schemas, references, invariants, generated/ up to date.

### 2 · REVIEW

CODEOWNERS requests suraj's review on every policy path; advisory today (0 approvals required; GAT-03 not started). Squash-merge.

### 3 · DELIVER

qqcfg deliver <repo> <checkout> writes the workflows into a product-repo checkout, opened as a PR; drift checks fail hand edits.

### 4 · PIN BUMP

Each tool sees the change only after its own infra-config pin is bumped.

**Repo settings are code too: edit gate's settings/github.toml, merge, then an org owner runs one command**

```
( mkdir -p ~/qq-apply && cd ~/qq-apply && rm -rf qq-gate && git clone -q https://github.com/quirq-ai/gate qq-gate && cd ./qq-gate && git checkout -q <COMMIT> && scripts/apply.sh )
```

It shows a dry run, writes only what you approve at its prompts, and is safe to re-run. Replace <COMMIT> with the gate commit you are given; details in gate's docs/apply-settings.md.

NAMING CONVENTIONS · AN ADOPTED PLAN

# The name says the area, the description says the job

An adopted org plan (quirq internal planning docs, not public), not something any repo enforces.

| Thing | Convention | Examples |
|---|---|---|
| Product repo | its brand name; a part is <product>-<part> | `xo-space, innernet` |
| One-repo area | the area word | `docs, research` |
| Multi-repo area | <area>-<thing>, the thing being its job | `infra—<thing> (new infra repos)` |
| The 13 infra repos | keep their names for v0; infra-* rename is optional, later | `gate, release` |
| Spelling | lowercase, words joined by -, no versions or years | |
| Description | <area>: <job in one line> | `docs: <its job in one line>` |
| Area and owner | org properties area, owner-bot, owner, stage; select by property, never by name | `area = infra` |
| Bots | one GitHub App per tool or area: quirq-<repo or area> | `quirq-gardener; rollers: quirqer` |
| qq's own files | the qq prefix | `qq CLI, .qq/, qq-*.yml` |
| Builders and checks | <repo>-<pipeline>; the job name is the required check | `xo-space-presubmit` |
| Work items | V<version>-<AREA>-<nn> | `V0-GAT-01, V1-PH-01` |
| Release refs | lkgr and channels/<channel> | `channels/canary` |

OWNERS AND BOTS

# Who decides, who acts

## People

suraj is the policy owner: policy paths, stable promotion and repo settings. Approvals are advisory today: qq-main requires 0, code-owner review is enforced only on toolchains, and gate.toml's change classes wait on V0-GAT-03.

org.toml names suraj owner of the org and all 13 infra repos (infra-config 310e326). Other config areas and the product repos in repos.toml still have empty owners lists (V0-ORG-02).

Every infra repo's CODEOWNERS names suraj on its policy and trust paths, with no catch-all.

```
qqcfg validate [--todos]
```

lists every empty owner list and rotation; --todos adds all open TODOs, 3 of them for v0.

## Agents and bots

Policy allows agents to land clean reverts, dependency rolls and docs alone. In v0 none of these land on their own (auto_land_repos is empty and roll auto-land is off). Canary promotion is the one thing agents do alone today.

One GitHub App per tool. Exists: quirqer (rollers). Not yet created: the gardener App (needed before exit test 4 can pass) and the release executor (product-repo refs and, later, depot release tags).

GitHub does not let a PR's author approve it: suraj can approve agent PRs (opened as quirq-gates), but suraj's own PRs need a second owner or a bot.

ADDING A NEW REPO

# Two PRs and a settings run; no tool code changes

The repo must be public in quirq-ai, named by the conventions, and use a known kind: python-service, pytest or node-app.

| Where | Change | Reviews |
|---|---|---|
| `infra-config repos.toml` | one [[repo]] block: source, kinds, manifest, channels, owners | suraj |
| `infra-config pipelines.toml` | presubmit and post-submit mirror; a canary builder if it ships canary (validate requires them) | suraj |
| `the new repo` | infra/repo.toml, workflows from qqcfg deliver, Dependabot files | repo owner |
| `gate settings/github.toml` | a [[repo]] entry, then the apply one-liner (org rulesets are off on GitHub Free) | suraj |
| GitHub | add to the rollers App install (gardener App once it exists); Dependabot on; auto-merge per gate settings | suraj (admin) |

**Follows by itself** once tools bump their pin: required checks, tree status, channels.

**Still a hand edit:** the test-pipelines scorecard and perf matrix list repos by name.

**New language:** a kind in kinds.toml, one recipes adapter, one toolchains build.

**Reviews are advisory today:** qq-main requires 0 approvals; code-owner review is enforced only on toolchains.

LIMITS TODAY · 2026-10-05

# The biggest gap is live proof

| Limit | Effect | Clears when |
|---|---|---|
| **Enforcement not proven live** | Repo rulesets applied, but no red-PR refusal recorded; org rulesets off on GitHub Free | Red-PR validation in both repos |
| **promotion-gate not required** | toolchains' promotion check stays advisory | Decided; turns on with the next settings run |
| **No gardener App or release executor** | No reverts opened; no lkgr or channel git refs; no depot v* tags | Gardener App: after the v0 code, needed before exit test 4 can pass. Executor: product-repo refs and, later, depot release tags |
| **Owners partly named** | Other config areas and product repos have none; GAT-03 waits | suraj names owners or a bot |
| **CI uses interim steps** | Builders use setup-python/node, not recipes; retry-then-compare off | Switching builders to the recipes adapters (open v0 work) |
| **Nothing lands alone** | auto_land_repos empty; roll auto-land off | Policy and gate settings PRs |
| **Canary ran once, by hand** | GitHub never fired the 06:17 slot, and lkgr ran 4 times overnight against a 10-minute schedule; a 07:37 UTC backstop routine now dispatches lkgr and the watchdog (first run green) | The unattended 7-day streak hasn't started yet (day 1 was by hand); a hold and a rollback drill |
| **Thin signals** | perf: one measurement per commit on shared runners; no PostHog yet | v1 (PostHog) |
| **Narrow platform** | GitHub only; toolchains for linux-x86_64 only | Launchpad (v2); more platforms |
| **Open values** | Canary hour, how the revert cap counts, monthly compute ceiling | suraj sets them |

WHAT'S NEXT

# Finish v0, gardener App included

## TO CLOSE v0

1. suraj's values (CFG-04) and the remaining owners (ORG-02), which unblock GAT-03

2. toolchains' promotion-gate: decided; turns on with the next settings run

3. Switch the generated product builders from interim commands to the recipes adapters (exit test 2)

4. Validate: a red PR refused in each product repo

5. Validate: 7 daily canaries with no human touch (streak not started; day 1 was by hand), a held bad canary, a rollback drill

6. The gardener App, then a planted break reverted automatically

## AFTER THE v0 CODE

The release executor, needed for the product repos' refs and, later, depot's release tags

Agents to land rolls and reverts alone, within caps (planned)

One-command, idempotent setup (V1-ORG-01)

Optional: rename infra repos to infra-*

Done since 2026-10-04: the settings run, owners for the org and infra repos, the 9123fa9 redelivery (xo-space #220, innernet #44) and the first canary (by hand).

SOURCES

# Where every status on these slides comes from

REPOS AT main, READ 2026-10-05
github.com/quirq-ai/<repo>/tree/<commit>

| | | | |
|---|---|---|---|
| depot | 741967c | sync | d97e2f7 |
| recipes | db6ce0a | toolchains | 44f8f95 |
| remote-build | 62b2f04 | infra-config | 310e326 |
| gate | c372136 | test-pipelines | a642399 |
| gardener | bf7d24d | rollers | 4285aa4 |
| perf | d1d9765 | release | 83ef917 |
| installer | d39f302 | xo-space | 14b21a4 |
| innernet | 8f383a3 | | |

**State branches**, fetched 2026-10-05: release `release-state` at d08a2fd (lkgr and canary pointers, channels.json, canary report 2026-10-05); gardener `tree-status` at 6733e4c; perf `perf-data` at 28628f3.

**Settings** as the repos record them (gate README and settings/github.toml); live GitHub settings were not read.

**Phases:** v0 plan in infra-config docs/v0.md (310e326). v1 and v2 are plans: quirq internal planning docs (not public).

**Naming:** an adopted org plan, 2026-10-04: quirq internal planning docs (not public).

**Write-ups:** quirq-ai/research, infra/output/report: the phases, v0 status, setup and build-and-run reports.